
Course Name: Computer Networks and Internet Protocol

Assignment 3 - Week 3 (Jan 2026)

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 10

Total mark: 10 X 1 = 10

QUESTION 1

Flow control is mainly implemented in

- a) Physical Layer
- b) Application Layer
- c) Transport Layer
- d) Session Layer

Correct Answer: (c)

Detailed Solution: Flow control is mainly a function of the Transport Layer.

QUESTION 2

What kind of sequence number does TCP use?

- a) byte-oriented sequence number
- b) packet-oriented sequence number
- c) Randomly generated fixed sequence numbers
- d) none of them

Correct Answer: (a)

Detailed solution: TCP uses byte-oriented sequence numbering to ensure reliable, ordered, and efficient data transmission over a network.

QUESTION 3

Which of the following services is NOT supported by the transport layer?

- a) End-to-end packet delivery
- b) Ordered packet delivery
- c) Reliable data delivery
- d) Forwarding the datagram from one hop to another hop in the network



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Correct Answer: (d)

Detailed solution: The transport layer provides services like End-to-end packet delivery, Ordered packet delivery, Reliable data delivery, Connection establishment, and Flow & Congestion control etc. The hop-to-hop network packet forwarding is supported by data link layer and network layer.

QUESTION 4

Transport layer is implemented in the Firmware of a computer system.

- a) True
- b) False

Correct Answer: (b)

Detailed Solution: Transport layer is implemented in the kernel.

QUESTION 5

A data of 40 bytes need to be delivered using TCP protocol but a sender can send a segment of maximum size 12 bytes only. Identify the sequence number of the last segment formed if the sequence number field uses 5 bits only? [Assume sequence number starts from 0]

- a) 1
- b) 4
- c) 12
- d) 31

Answer: (b)

Detailed solution:

TCP uses byte sequencing. It treats data as a continuous stream of bytes rather than discrete packets. Since the sequence number field uses 5 bits only, it produces sequence numbers ranging from 0 to 31.

The first segment of 12 bytes will use the sequence number range (0-11),

Next segment of 12 bytes will use the sequence number range (12-23),

Next segment of 12 bytes will use the sequence number range (24-31) & (0-3), and

The last segment of 4 bytes will use the sequence number range (4-7).

The sequence number in a TCP segment header refers to the first byte of data in that segment relative to the start of the stream.



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QUESTION 6

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Timestamping a network packet is necessary for

- (i) Enabling time synchronization across router in network layer
 - (ii) For defining packet lifetime.
- a) Only (i)
 - b) Only (ii)
 - c) Both (i) and (ii)
 - d) None of above

Correct Answer: (c)

Detailed Solution: Timestamping is required for both defining packet lifetime and ensuring time synchronization across router in network layer.

QUESTION 7

Determine whether the following information is True or False?

“During the three-way handshaking of connection establishment procedure (from Host 1 to Host 2), Delayed duplicate SYN can be handled by TCP at Host 2 simply by ignoring it, as the sequence number is invalid”

- a) True
- b) False

Correct Answer: (a)

Detailed solution: The TCP 3-Way Handshake is a fundamental process that establishes a reliable connection between two devices over a TCP/IP network. During the handshake, the client and server exchange initial sequence numbers. It uses sequence numbers to identify and validate the legitimacy of connection requests. A delayed duplicate SYN will have an outdated sequence number that doesn't match the current connection state at Host 2 and can be ignored as invalid sequence number.

QUESTION 8

TCP instance uses a sliding window protocol in the noisy channel. A timeout occurs due to the loss of a segment (or the acknowledgement). Mark the RIGHT statement among the following-

- a) In the case of Go-Back-N ARQ, if any segment of the sliding window is lost, all the segments of the sliding window are retransmitted while in Selective Repeat ARQ, only the lost packets are selectively transmitted.
- b) In the case of Go-Back-N ARQ, if any segment of the sliding window is lost, only the lost packets of the sliding window are selectively retransmitted while in Selective Repeat ARQ, all packets of the sliding window are retransmitted.
- c) In both cases, Go-Back-N ARQ and Selective Repeat ARQ, all packets of the sliding window are retransmitted.
- d) In the both cases, Go-Back-N ARQ and Selective Repeat ARQ, only lost packets of the sliding window are selectively retransmitted.

Correct Answer: (a)

Detailed solution: In the case of Go-Back-N ARQ, if any segment of the sliding window is lost, all the segments of the corresponding sliding window are retransmitted while in Selective Repeat ARQ, only the lost packets are selectively transmitted.

QUESTION 9

What is known as outstanding frames?

- a) Frames that are yet to be transmitted
- b) Frames that have been transmitted, but not yet acknowledged
- c) Acknowledged frames
- d) None of the above

Correct Answer: (b)

Detailed solution: Frames that have been transmitted, but not yet acknowledged are known as outstanding frames.

QUESTION 10

A client and a server establish a connection using TCP's three-way handshake. The following details are given:

- Initial sequence number of the client: 100 and initial sequence number of the server: 300.
- RTT (Round Trip Time): 80ms.
- Each TCP segment carries 40 bytes of header information.
- The server sends a SYN-ACK segment, and the acknowledgement (ACK) segment from the client is delayed due to congestion, taking 100ms to reach the server.

How much time does it take to establish the connection under such a scenario?

- a) 100ms
- b) 280ms
- c) 180ms
- d) 260ms

Correct Answer: (c)

Detailed solution: In three way handshaking, the connection is established after three steps. In first step, sender sends a connection request SYN packet to the server. Server responds to client with SYN+ACK packet. Finally, sender sends a ACK packet to the server. Since $RTT = 80ms$ so for one way packet transmission will take $40ms$. Since, there is a delay during transmission of ACK packet from sender to server due to congestion and it takes greater time i.e. $100ms$. So, the total time needed to complete this connection procedure will be (time taken for SYN packet + time taken for SYN+ACK packet + time taken for ACK packet) i.e. $(80/2ms + 80/2ms + 100ms = 180ms)$.
